

An Analysis of Mixture model

Mixture model

Mixture model -- Part 1

In statistics, a mixture model is a probabilistic model for representing the presence of subpopulations within an overall population, without requiring that an observed data set should identify the sub-population to which an individual observation belongs. Formally a mixture model corresponds to the mixture distribution that represents the probability distribution of observations in the overall population. However, while problems associated with "mixture distributions" relate to deriving the properties of the overall population from those of the sub-populations, "mixture models" are used to make statistical inferences about the properties of the sub-populations given only observations on the pooled population, without sub-population identity information. Mixture models are used for clustering, under the name model-based clustering, and also for density estimation. Mixture models should not be confused with models for compositional data, i.e., data whose components are constrained to sum to a constant value (1, 100%, etc.). However, compositional models can be thought of as mixture models, where members of the population are sampled at random. Conversely, mixture models can be thought of as compositional models, where the total size reading population has been normalized to 1.

Mixture model -- Part 2

$\theta_{i=1 \dots K, j=1 \dots V} : \{\text{displaystyle } \theta_{i=1 \dots K, j=1 \dots V}\}$ probability for component i $\{\text{displaystyle } i\}$ of observing item j $\{\text{displaystyle } j\}$

Mixture model -- Part 3

and $p_2 = 1 - p_0 - p_1$. Clearly, given p_0 and p_1 , it is not possible to determine the above mixture model uniquely, as there are three parameters (π, θ_1, θ_2) to be determined.

Mixture model -- Part 4

Moment matching The method of moment matching is one of the oldest techniques for determining the mixture parameters dating back to Karl Pearson's seminal work of 1894. In this approach the parameters of the mixture are determined such that the composite distribution has moments matching some given value. In many instances extraction of solutions to the moment equations may present non-trivial algebraic or computational problems. Moreover, numerical analysis by Day has indicated that such methods may be inefficient compared to EM. Nonetheless, there has been renewed interest in this method, e.g., Craigmille and Titterton (1998) and Wang. McWilliam and Loh (2009) consider the characterisation of a hyper-cuboid normal mixture

copula in large dimensional systems for which EM would be computationally prohibitive. Here a pattern analysis routine is used to generate multivariate tail-dependencies consistent with a set of univariate and (in some sense) bivariate moments. The performance of this method is then evaluated using equity log-return data with Kolmogorov–Smirnov test statistics suggesting a good descriptive fit.

Mixture model -- Part 5

In image processing and computer vision, traditional image segmentation models often assign to one pixel only one exclusive pattern. In fuzzy or soft segmentation, any pattern can have certain "ownership" over any single pixel. If the patterns are Gaussian, fuzzy segmentation naturally results in Gaussian mixtures. Combined with other analytic or geometric tools (e.g., phase transitions over diffusive boundaries), such spatially regularized mixture models could lead to more realistic and computationally efficient segmentation methods.

Mixture model -- Part 6

K is said to be identifiable if all its members are unique, that is, given two members p and p' in K , being mixtures of k distributions and k' distributions respectively in J , we have $p = p'$ if and only if, first of all, $k = k'$ and secondly we can reorder the summations such that $a_i = a'_i$ and $f_i = f'_i$ for all i .

Mixture model -- Part 7

$\theta_{i=1 \dots K} : \{\text{displaystyle } \theta_{i=1 \dots K}\}$ vector of dimension V , $\{\text{displaystyle } V\}$ composed of θ_i , $1 \dots V$; $\{\text{displaystyle } \theta_{i=1 \dots V}\}$ must sum to 1

Mixture model -- Part 8

$p(\theta) = \sum_{i=1}^K \phi_i N(\mu_i, \Sigma_i) \{\text{displaystyle } p(\theta) = \sum_{i=1}^K \phi_i N(\mu_i, \Sigma_i)\}$

Mixture model -- Part 9

$\Sigma_{s(j+1)} = \sum_{t=1}^N h_{s(j)}(t) [x(t) - \mu_{s(j+1)}] [x(t) - \mu_{s(j+1)}]^T \quad \sum_{t=1}^N h_{s(j)}(t) \{\text{displaystyle } \Sigma_{s(j+1)} = \sum_{t=1}^N h_{s(j)}(t) [x(t) - \mu_{s(j+1)}] [x(t) - \mu_{s(j+1)}]^T \quad \sum_{t=1}^N h_{s(j)}(t)\}$

Mixture model -- Part 10

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$$p_0 = \pi(1 - \theta_1)^2 + (1 - \pi)(1 - \theta_2)^2 \quad p_1 = 2\pi\theta_1(1 - \theta_1) + 2(1 - \pi)\theta_2(1 - \theta_2) \quad \{\displaystyle \begin{aligned} p_0 &= \pi \left(1 - \theta_1\right)^2 + \left(1 - \pi\right) \left(1 - \theta_2\right)^2 \\ p_1 &= 2\pi \theta_1 \left(1 - \theta_1\right) + 2 \left(1 - \pi\right) \theta_2 \left(1 - \theta_2\right) \end{aligned} \}$$

Lock and more recently minimum message length (MML) techniques such as Figueiredo and Jain and to some extent the moment matching pattern analysis routines suggested by McWilliam and Loh (2009).

Mixture model -- Part 11

Point set registration Probabilistic mixture models such as Gaussian mixture models (GMM) are used to resolve point set registration problems in image processing and computer vision fields. For pair-wise point set registration, one point set is regarded as the centroids of mixture models, and the other point set is regarded as data points (observations). State-of-the-art methods are e.g. coherent point drift (CPD) and Student's t-distribution mixture models (TMM). The result of recent research demonstrate the superiority of hybrid mixture models (e.g. combining Student's t-distribution and Watson distribution/Bingham distribution to model spatial positions and axes orientations separately) compare to CPD and TMM, in terms of inherent robustness, accuracy and discriminative capacity.

Mixture model -- Part 12

Parameter estimation and system identification Parametric mixture models are often used when we know the distribution Y and we can sample from X , but we would like to determine the μ_i and θ_i values. Such situations can arise in studies in which we sample from a population that is composed of several distinct subpopulations. It is common to think of probability mixture modeling as a missing data problem. One way to understand this is to assume that the data points under consideration have "membership" in one of the distributions we are using to model the data. When we start, this membership is unknown, or missing. The job of estimation is to devise appropriate parameters for the model functions we choose, with the connection to the data points being represented as their membership in the individual model distributions. A variety of approaches to the problem of mixture decomposition have been proposed, many of which focus on maximum likelihood methods such as expectation maximization (EM) or maximum a posteriori estimation (MAP). Generally these methods consider separately the questions of system identification and parameter estimation; methods to determine the number and functional form of components within a mixture are distinguished from methods to estimate the corresponding parameter values. Some notable departures are the graphical methods as outlined in Tarter and